

Applied Optimization - Ex 3

QCQP to SOCP

$$1.) (\Leftrightarrow) \left\| \frac{1 + (b^T x + c)}{Ax} \right\|_2 \leq \frac{1 - (b^T x + c)}{2}$$

$$\|Ax\|_2^2 + \left(\frac{1+u}{2}\right)^2 \leq \left(\frac{1-u}{2}\right)^2$$

$$\|Ax\|_2^2 + \frac{1+2u+u^2}{4} \leq \frac{1-2u+u^2}{4}$$

$$\|Ax\|_2^2 + \frac{4u}{4} \leq 0$$

$$x^T A^T A x + b^T x + c \leq 0$$

$$(\Rightarrow) x^T A^T A x + b^T x + c \leq 0$$

$$\|Ax\|_2^2 + u \leq 0$$

reverse algebra above

$$\left\| \frac{1 + (b^T x + c)}{Ax} \right\|_2 \leq \frac{1 - (b^T x + c)}{2}$$

$$2.) \text{ minimize } x_1^2 + 4x_1x_2 + 4x_2^2 = (x_1 + 2x_2)^2$$

$$\bullet \text{ s.t. } \begin{cases} 9x_1^2 + 16x_2^2 \leq 25 \\ x_1 - x_2 = 1 \end{cases} \text{ already in SOCP form}$$

$\Rightarrow$ epigraph transform

$$\begin{aligned} &\text{minimize } t \\ &\text{s.t. } x_1^2 + 4x_1x_2 + 4x_2^2 \leq t \quad (\text{quadratic inequality constraint}) \end{aligned}$$

$\Rightarrow$ algebra

$$x_1^2 + 4x_1x_2 + 4x_2^2 = x^T H x = x^T A^T A x = \|Ax\|_2^2$$

$\hookrightarrow$ for some A

$$\Rightarrow \|Ax\|_2 \leq \sqrt{t}$$

not affine

$$\Rightarrow \|Ax\|_2 \leq S, \quad S^2 \leq t$$

Not in SOCP form

$$\Rightarrow \|Ax\|_2 \leq S, \quad (2S, 1, t) \in Q_r^3$$

rotated second order cone

(let $u = b^T x + c$, and square both sides)

$$(\text{Note: } \left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\|_2^2 = \left\| \begin{bmatrix} x_2 \\ x_1 \end{bmatrix} \right\|_2^2 + x_1^2)$$

Final problem

$$\begin{aligned} &\text{minimize } t \\ &\text{s.t. } \left\| \begin{bmatrix} 3x_1 \\ 4x_2 \end{bmatrix} \right\|_2 \leq 5 \end{aligned}$$

$$x_1 - x_2 = 1$$

$$\|Ax\|_2 \leq S$$

$$(2S, 1, t) \in Q_r^3$$

Linear Programming

$$\text{minimize } \|(2x_1 + 3x_2, -3x_1)^T\|_\infty$$

$$\text{s.t. } |x_1 - 2x_2| \leq 3$$

$$(1) \text{ Note } \|x\|_\infty = \max\{|x_1|, \dots, |x_n|\}$$

$$\text{minimize } t$$

$$x, t$$

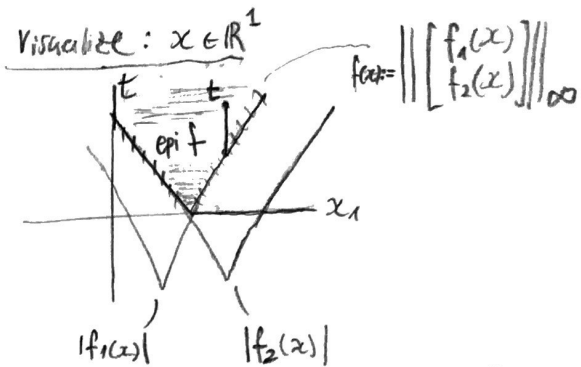
$$\text{s.t. } |2x_1 + 3x_2| \leq t$$

$$|-3x_1| \leq t$$

$$|x_1 - 2x_2| \leq 3$$

$$-(x_1 - 2x_2) \leq 3$$

epigraph transform.



$$\text{minimize } t$$

$$x, t$$

$$\text{s.t. } \pm(2x_1 + 3x_2) \leq t$$

$$\pm(-3x_1) \leq t$$

$$\pm(x_1 - 2x_2) \leq 3$$

6 constraints total (2 for each $\pm$)

(2) (a) All vars in $\mathbb{R}_+$

$$\text{minimize } t$$

$$x, t$$

$$\text{s.t. } \pm(2(x_1^+ - x_1^-) + 3(x_2^+ - x_2^-)) \leq t$$

$$\pm(-3(x_1^+ - x_1^-)) \leq t$$

$$\pm(x_1^+ - x_1^-) - 2(x_2^+ - x_2^-) \leq 3$$

$$x_1^+, x_1^-, x_2^+, x_2^- \geq 0$$

(b) Remove general inequality constraints

$$\text{minimize } t$$

$$x, t$$

$$\text{s.t. } 2(x_1^+ - x_1^-) + 3(x_2^+ - x_2^-) - t + s_1 = 0$$

$$-(-3(x_1^+ - x_1^-)) - t + s_2 = 0$$

$$-3(x_1^+ - x_1^-) - t + s_3 = 0$$

$$3(x_1^+ - x_1^-) - t + s_4 = 0$$

$$(x_1^+ - x_1^-) - 2(x_2^+ - x_2^-) - 3 + s_5 = 0$$

$$-(x_1^+ - x_1^-) - 3 + s_6 = 0$$

$$x_1^+, x_1^-, x_2^+, x_2^-, s_1, \dots, s_6 \geq 0$$